

2.9.1 T-connections are generally to be made by fillet welds on both sides of the abutting plate, the dimensions and spacing of which are shown in [Figure 13.2.1 Weld dimensions and types](#). Where the connection is highly stressed, deep penetration or full penetration welding may be required. Where full penetration welding is required, the abutting plate may be required to be bevelled.

- 2.9.2 Where an approved deep penetration procedure is used, the fillet leg length calculated may be reduced by 15 per cent provided that the manufacturer is able to meet the following requirements:
- a. Use of a welding consumable approved for deep penetration welding in accordance with [Ch 11 Approval of Welding Consumables](#) for either the ‘p’ or ‘T’ techniques.
 - b. Demonstrations by way of production weld testing that the minimum required penetration depths (i.e. throat thicknesses) are maintained. This is to be documented on a monthly basis by the manufacturer and be available to the Surveyor.

- 2.9.3 The calculated fillet leg length may be reduced by 20 per cent, provided that in addition to the requirements of [Ch 13, 2.9 Fillet welds 2.9.2.\(a\)](#) and [Ch 13, 2.9 Fillet welds 2.9.2.\(b\)](#), the manufacturer is able to consistently meet the following additional requirements:
- a. The documentation required in [Ch 13, 2.9 Fillet welds 2.9.2.\(b\)](#) is to be completed and made available to the Surveyor upon request on a weekly basis.
 - b. Suitable process selection confirmed by satisfactory welding procedure tests covering both minimum and maximum root gaps.

2.9.4 Where intermittent welding is used, the welding is to be made continuous in way of brackets, lugs and scallops and at orthogonal connections with other members.

2.10 Post-weld heat treatment

- 2.10.1 This section determines the requirements for post-weld stress relief heat treatment when applied to ship structure and associated machinery.
- 2.10.2 Post-weld stress relief heat treatment is applied to improve resistance to brittle fracture or fatigue performance. It is to be applied when the thickness limits stated in [Table 13.2.4 Post-weld stress relief heat treatment thickness limits](#) are exceeded.

Table 13.2.4 Post-weld stress relief heat treatment thickness limits

Typical Application	Load Conditions (<i>See Note 1</i>)	Material Grade (<i>See Note 2</i>)	Material Thickness Limit (mm) (<i>See Note 3</i>)
Ship structure	Fatigue non-critical / critical	Normal strength (A - E), Higher strength (H27S - H40, [Excluding EH47])	150
Ship structure	Fatigue non-critical / critical	Higher strength (EH47)	100
Machinery	Fatigue non-critical	High strength (H42 - H69)	140
Machinery	Fatigue critical	High strength (H42 - H46)	100
Machinery	Fatigue critical	High strength (H50 - H69)	65
Ship structure / Machinery	Any	Other material grades	Subject to special consideration

- Note 1.** Fatigue analysis shall be approved by design appraisal according to relevant rules for each application.
- a. Fatigue non-critical – Design assessment confirms that there are cyclic stresses but the fatigue life is reasonably greater than the design fatigue life and it is anticipated that fatigue crack initiation and propagation are unlikely to occur.
 - b. Fatigue critical – Design assessment confirms that there are cyclic stresses and the estimated fatigue life meets the design requirements but it is not significantly higher. It is anticipated that fatigue crack initiation and propagation are likely to occur.
- Note 2.** Where steel grades based on national and international standards are specially agreed for construction, they are to be procured from LR approved works which hold current approval for the LR equivalent grade and the same delivery condition as the steel to be procured.

Note 3. For all applications where material thickness is greater than 65 mm (or greater than 100 mm for EH47), 100 per cent surface and volumetric non-destructive examination of welds is required.

2.10.3 Post-weld heat treatment is to be applied to the following types of welded construction:

- a. Welding of steel castings where the thickness of the casting at the weld exceeds 30 mm, except where castings are directly welded to the hull structure.
- b. Engine bedplates except for engine types where the bedplate as a whole is not subjected to direct loading from the cylinder pressure. For these types, only the transverse girder assemblies need to be stress relieved.
- c. Welding of gear wheels.
- d. Welding of gear cases associated with main or auxiliary engines, see [Pt 5 Main and Auxiliary Machinery](#).

2.10.4 Consideration is to be given to applying post-weld heat treatment for all thicknesses of complicated weld joints where there are high stress concentrations.

2.10.5 Where required, heat treatment is to be performed in accordance with the requirements specified in [Ch 13, 1.16 Post-weld heat treatment](#) and [Table 13.4.3 Post-weld soak temperatures and times](#).

2.10.6 Special consideration may be given to omit the required post-weld heat treatment. Evaluation is to be based on critical engineering assessment involving fracture mechanics testing and proposals are to be submitted which include full details of the application, materials, welding procedures, inspection procedures, design temperature and stresses, fatigue loads and cycles. Evidence will be required to demonstrate that the inspection techniques and procedures to be employed are able to detect flaws to the sizes and tolerances (of length, through-wall height and through-wall position), as determined from the fracture mechanics (and or fatigue) calculations. Alternative procedures for the omission of post weld heat treatment will be subject to special consideration.

2.11 Tolerances

2.11.1 Tolerances after welding are to be in accordance with the relevant Part of the Rules.

2.11.2 Distortion which has resulted from welding may be corrected by spot heating in accordance with [Ch 13, 1.14 Rectification of distortion](#).

2.12 Non-Destructive Examination of steel welds

2.12.1 All finished welds are to be sound and free from cracks and substantially free from lack of fusion, incomplete penetration, porosity and slag. The surfaces of welds are to be reasonably smooth and substantially free from undercut and overlap. Care is to be taken to ensure that the specified dimensions of welds have been achieved and that both excessive reinforcement and under-fill of welds is avoided.

2.12.2 Welds forming part of the hull and superstructure may be coated with a thin layer of protective primer prior to inspection provided it does not interfere with inspection and is removed, if required by the Surveyor, for closer interpretation of possible defective areas.

2.12.3 All welds are to be visually inspected by personnel designated by the builder. Visual inspection of all welds may be supplemented by other non-destructive examination techniques in cases of unclear interpretation, as considered necessary. The acceptance criteria for visual testing are given in [Table 13.2.5 Acceptance criteria for visual testing, magnetic particle and liquid penetrant testing](#).

Table 13.2.5 Acceptance criteria for visual testing, magnetic particle and liquid penetrant testing

Surface discontinuity	Classification according to ISO 6520-1	Acceptance criteria
Crack	100	Not accepted
Lack of fusion	401	Not accepted
Incomplete root penetration in butt joints welded from one side	4021	Not accepted
Surface pore	2017	Visual inspection
		Thickness (<i>t</i>)

		0,5 mm < $t \leq$ 3,0 mm	$t >$ 3,0 mm
		Not permitted	Butt welds: $d \leq 0,2 t$ (max of 2,0 mm) Fillet welds: $d \leq 0,2 a$ (max of 2,0 mm) <i>See</i> Notes 4, 5 & 6
		Liquid penetration inspection	
		Single pore indication diameter $d \leq 6$ mm <i>see</i> Notes 1, 2, 3 & 4	
		Magnetic particle inspection	
		Single pore diameter $d \leq 3$ mm <i>see</i> Notes 1, 3 & 4	
		d = major axis of dimension	
Undercut	5011 (Continuous)	Thickness (t)	
	5012 (Intermittent)	0,5 mm < $t \leq$ 3,0 mm	$t >$ 3,0 mm
		Short imperfections only <i>see</i> Notes 7 & 8: $h \leq 0,1 t$ <i>see</i> Note 9	Short imperfections only <i>see</i> Notes 7 & 8: $h \leq 0,1 t$ (max 0,5 mm) <i>see</i> Note 9
		Smooth transition to parent material is required and imperfection is not to be regarded as systematic.	
Note 1. A pore is defined as an indication having a length less than or equal to three times its width. Note 2. A penetrant indication refers to the size of the bleed out from the discontinuity resulting in the indication. Note 3. Indications that are approximately in line, which are separated by less than the length of the smaller indication, are to be considered as a single indication. Note 4. d = diameter. Note 5. t = thickness of thinner material. Note 6. a = throat thickness. Note 7. For either continuous or intermittent undercut, only short imperfections are allowed. Note 8. The definitions of short imperfections are as follows: - For welds 100 mm long or longer: Imperfections whose total length is not greater than 25 mm in the 100 mm of the weld which contains the greatest number of imperfections. - For welds less than 100 mm long: Imperfections whose total length is not greater than 25 per cent of the length of the weld. Note 9. h = height or width of imperfection.			

2.12.4 In addition to visual inspection, where required by either LR Rules, the NDE checkpoint plan, the contract inspection and test plan, or as warranted for further testing either by the manufacturer or the surveyor, welded joints are to be examined using any one or a combination of ultrasonic, radiographic, magnetic particle, eddy current, dye penetrant or other acceptable methods appropriate to the configuration of the weld.

2.12.5 The method to be used for the volumetric examinations of welds is the responsibility of the builder; however, the following technical considerations shall be noted for the choice concerning the selected method:

- a. For full penetration butt welds, advanced NDE (ANDE) methods may be used in lieu of (or complementary to) existing ultrasonic or radiographic testing methods. These methods may additionally be used on other weld configurations, with some limitations, as specified in [Table 13.2.9 Applicable methods for testing of materials and weld joints](#).
- b. Radiography (using film or RT-D methods) may be used for the examination of welds for any thickness range, as applicable to the penetrating capability of the radiation energy and the radiation source, and within any limits as identified in the procedure in order to achieve the specified quality level. The applicable material and joint types are given in [Table 13.2.9 Applicable methods for testing of materials and weld joints](#).
- c. Ultrasonic testing may be used for the examination of welds, generally for 8 mm thickness or greater, and advanced methods (such as PAUT or TOFD) for thicknesses of 6 mm or greater (as appropriate). The applicable material and joint types are given in [Table 13.2.9 Applicable methods for testing of materials and weld joints](#).
- d. Where there is a requirement for enhanced NDE acceptance criteria to be applied to thick plate sections in the hatch coaming region of container ships, as per the Measure 3 requirement in [Table 8.2.1 Chemical composition, percentage](#), as described in [Pt 4, Ch 8, 2.3 Requirements for use of thick steel plates 2.3.10](#) of the [LR-RU-001 Rules and Regulations for the Classification of Ships, July 2024](#), incorporating [Notice No. 1, 2 & 3](#), the UT and PAUT acceptance criteria are to be derived from the ShipRight Procedure for the *Use of Enhanced NDE in Container Ships*. These derived acceptance criteria are project specific, and the acceptance criteria stated in [Table 13.2.7 Acceptance criteria for ultrasonic and phased array testing](#) are not applicable.

2.12.6 The acceptance criteria for volumetric weld testing as applied to the appropriate methods are given in the following tables:

- a. Radiographic testing (including RT-D): [Table 13.2.6 Acceptance criteria for radiographic testing](#)
- b. Ultrasonic testing and PAUT (based on length and amplitude of indications): in [Table 13.2.7 Acceptance criteria for ultrasonic and phased array testing](#)
- c. TOFD testing (based on length and height of indications): [Table 13.2.8 Acceptance criteria for TOFD testing](#)¹. *See also* [Figure 13.2.5 General scheme for acceptance conditions](#) for the general approach to acceptance/rejection and interpretation of signal parameters. Other acceptance criteria, including project specific acceptance criteria, are to be specially agreed with LR.

Table 13.2.6 Acceptance criteria for radiographic testing

Discontinuity	Classification according to ISO 6520-1	Acceptance criteria
Crack	100	Not permitted
Lack of fusion	401	Acceptable up to but only intermittently and not breaking the surface, $\sum l \leq 25$ mm, $L = 100$ mm. and $l \leq 0,4 s$, maximum 4 mm <i>See</i> Notes 1, 2. 5 & 9
Lack of penetration	402	Not permitted
Slag inclusions, Flux inclusions, & Oxide inclusions	301, 302 & 303	$h < 0,3 s$ (max 3,0 mm) $\sum l \leq s$ (max 50 mm) $L = 100$ mm <i>See</i> Notes 1, 5 & 8.
Porosity & Gas pore (Single Layer)	2011 & 2012	$A \leq 1,5 \%$ $d \leq 0,3 s$ (max 4,0 mm) $L = 100$ mm <i>See</i> Notes 1, 3, 4, 5, 6 & 7.
Porosity & Gas pore (Multi-Layer)	2011 & 2012	$A \leq 3,0 \%$ $d \leq 0,3 s$ (max 4,0 mm) $L = 100$ mm <i>See</i> Notes 1, 3, 5, 6 & 7.
Linear porosity	2014	$l \leq s$, max 50 mm

		$d \leq 0,3 s$ (max 3,0 mm) $L = 100$ mm <i>See Notes 1, 3, 5, 6, 7 & 10.</i>
Clustered (localised) porosity	2013	$dA \leq W_p$ (max 20 mm) $L = 100$ mm <i>See Notes 1, 3, 7, 10 & 11.</i>
Elongated cavity & wormholes	2015 & 2016	$h < 0,3 s$ (max 3,0 mm) $\sum l \leq s$ (max 50 mm) $L = 100$ mm <i>See Notes 1, 4, 7, 8 & 10.</i>
Shrinkage cavity (other than crater pipes)	202	Not permitted
Crater pipe	2024	Not permitted
Metallic inclusions other than copper	304	$l \leq 0,3 s$ (max 3,0 mm) <i>See Notes 2 & 5</i>
Copper inclusions	3042	Not permitted

- Note 1.** L = any 100 mm testing length within the radiograph.
- Note 2.** l = Length of indication (mm).
- Note 3.** A = Sum of projected areas of indications related to $L \times W_p$, in %.
- Note 4.** h = Width of indication, the width or height of surface imperfection (mm).
- Note 5.** s = Nominal butt weld thickness (mm).
- Note 6.** d = Diameter of pore (mm).
- Note 7.** W_p = Width of weld (mm).
- Note 8.** $\sum l$ = Summary length of imperfections within L (mm).
- Note 9.** If the length of the weld is below 100 mm then the maximum length of indications is not to exceed 25% of that weld.
- Note 10.** For details regarding the sum of acceptable areas for porosity, see [Figure 13.2.2 Sum of acceptable areas for radiography](#).
- Note 11.** d_A = Diameter of pore envelope